

Brolga Guardians Part 1: Multi-Period Supply Chain Optimisation Model

June 25, 2026

Section A – Report to your boss

Sets

S	Burn sites
W	Warehouses $\subseteq S$
E	Edges (Roads)
T	Years
R	Resources

Data

c_{sr}	Cost (\$/L) of resource $r \in R$ for site $s \in W$.
a_{sr}	Availability (L/year) of resource $r \in R$ for site $s \in W$.
d_{srt}	Demand (L) for resource $r \in R$ for burn site $s \in S$ in year $t \in T$.
l_e	Length (km) of edge $e \in E$.
f_e, t_e	"From" and "to" (site ID) for edge $e \in E$.
Cap_e	Capacity (L of resources) of edge $e \in E$.
$TransCost_t$	Transportation cost (\$/L/KM) for transporting resources for year $t \in T$.
$EvapRate_r$	Evaporation rate (%/km) for resource $r \in R$.
$MaxStore$	Maximum storage (L) of resources at non-warehouse sites.

Variables

x_{srt}	Volume (L) of resource $r \in R$ to purchase from site $s \in S$ for year $t \in T$.
y_{ert}	Flow (L) of resource $r \in R$ along edge $e \in E$ in year $t \in T$.
z_{srt}	Volume (L) of resource $r \in R$ to store at site $s \in S$ for year $t \in T$.

Objective

$$\min \left\{ \sum_{s \in W} \sum_{r \in R} \sum_{t \in T} c_{sr} \cdot x_{srt} + \sum_{e \in E} \sum_{r \in R} \sum_{t \in T} TransCost_t \cdot l_e \cdot y_{ert} \right\}$$

Constraints

$$\text{Road Capacity: } \sum_{r \in R} y_{ert} \leq Cap_e \quad \forall e \in E, \forall t \in T \quad (1)$$

$$\text{Resource availability: } x_{srt} \leq a_{sr} \quad \forall s \in W, \forall r \in R, \forall t \in T \quad (2)$$

$$x_{srt} \leq 0 \quad \forall s \in S \setminus W, \forall r \in R, \forall t \in T \quad (3)$$

$$\text{Storage limit: } \sum_{r \in R} z_{srt} \leq MaxStore \quad \forall s \in S \setminus W, \forall t \in T \quad (4)$$

$$\sum_{r \in R} z_{srt} \leq 0 \quad \forall s \in W, \forall t \in T \quad (5)$$

Flow Balance:

$$x_{srt} + z_{srt-1} + (1 - EvapRate_r \cdot l_e) \sum_{\substack{e \in E \\ \text{s.t. } t_e = s}} y_{ert} = d_{srt} + z_{srt} + \sum_{\substack{e \in E \\ \text{s.t. } f_e = s}} y_{ert} \quad \forall s \in S, \forall r \in R, \forall t \in T, t > 0 \quad (6)$$

$$x_{srt} + (1 - EvapRate_r \cdot l_e) \sum_{\substack{e \in E \\ \text{s.t. } t_e = s}} y_{ert} = d_{srt} + z_{srt} + \sum_{\substack{e \in E \\ \text{s.t. } f_e = s}} y_{ert} \quad \forall s \in S, \forall r \in R, t = 0 \quad (7)$$

$$\text{Non-negativity: } x_{srt} \geq 0 \quad \forall s \in S, \forall r \in R, \forall t \in T \quad (8)$$

$$y_{ert} \geq 0 \quad \forall e \in E, \forall r \in R, \forall t \in T \quad (9)$$

$$z_{srt} \geq 0 \quad \forall s \in S, \forall r \in R, \forall t \in T \quad (10)$$

Constraint (1) ensures that the amount of resources transported on each road does not exceed the road capacity limit.

Constraints (2-3) ensure that the amount of resources purchased from each warehouse must not exceed the availability of the resource at each warehouse.

Constraint (4-5) ensures that the resources stored at each non-warehouse site do not exceed the storage limit.

Constraint (6-7) defines the flow balance so that enough resources are purchased to satisfy all sites' demands and storage over the years. (6) specifically handles with the first year.

Constraints (8-10) enforce non-negativity requirements.

Section B – Report to Brolga Guardians

To the stakeholders at Brolga Guardians,

The following sections are our responses to your communications in order of receipt, with the results of our analysis.

Working Assumptions

During the course of this analysis, the following assumptions have been made:

- Suppliers' price of fuel and suppressant is outside the control of Brolga Guardians (the client has sought out the most competitive prices)
- Does not take into account unforeseen traffic delays or road blockages that result in rerouting of transport vehicles

Brolga is able to modify/take measures to improve:

- Warehouse capacities
- Road conditions
- Reducing the evaporation rate

Communication 1

Your initial request was to optimise the delivery of burn fuel from four warehouse locations to the burn sites, as detailed in your initial communication.

Based on the data provided, Brolga Guardians can optimise fuel purchasing and transport expenditure as per the following:

- Purchasing costs: \$70,609.48
- Transport costs: \$820,197.70
- Total costs: \$890,807.18

Distribution of fuel purchases is detailed in Table [1](#).

Warehouse	Volume in L
A	1,194.0
B	3,100.0
C	2,100.0
D	2,300.0

Table 1: Optimal distribution of fuel to minimise cost. Warehouses operating at full capacity are shown in red.

Warehouses at maximum capacity

As you can see in the cost distribution, the biggest expense is transportation, representing 92% of the costs. One way to optimise it further is to expand the capacity of warehouses that are closer to burn sites. In our optimal solution, warehouses B, C and D are fully utilised, but if warehouse C's capacity were increased by 1,240 L to a total of 3,340 L, the optimal solution would be:

- Purchasing costs: \$75,390.78
- Transport costs: \$775,332.05
- Total costs: \$850,722.83

Distribution of fuel purchases is detailed in Table 2.

Warehouse	Volume in L
A	1,078.0
B	2,742.0
C	3,336.0
D	1,538.0

Table 2: Optimal distribution of fuel to minimise cost after increasing warehouse C's capacity.

This would mean a reduction of \$40,084.35 (4.5%) in total expenditure, particularly impacted by savings in transportation, from \$820,197.7 to \$775,332.05.

Communication 2

Your request was updated to allow for maximum road capacities during transport, due to poor road conditions in some areas, as outlined in your second communication. The optimised costs of fuel delivery from the four warehouse locations to the burn sites are as follows:

- Purchasing costs: \$71,314.21
- Transport costs: \$832,397.04
- Total costs: \$903,711.25

Distribution of fuel purchases is detailed in Table 3.

Warehouse	Volume in L
A	1,333.0
B	3,100.0
C	2,100.0
D	2,161.0

Table 3: Optimal distribution of fuel to minimise cost. Warehouses operating at full capacity are shown in red.

Warehouses at maximum capacity

Again, the biggest expense is transportation, representing 92% of the costs. One way to optimise it further is to expand warehouse C's capacity. In this setting (considering road limits), increasing it by 1,000 L to a total of 3,100 L, would give the optimal cost:

- Purchasing costs: \$75,118.20
- Transport costs: \$790,881.27
- Total costs: \$865,999.47

This would mean a reduction of \$37,711.78 (4.17%) in total expenditure, particularly impacted by reducing transportation costs from \$832,397.04 to \$790,881.27. Distribution of fuel purchases is detailed in Table 4.

Roads at maximum capacity

The roads connecting the sites (45, 41), (41, 30), (4, 51), (53, 42), (5, 23) are operating at maximum capacity. Another way to further optimise costs would be to increase these roads' capacities.

Warehouse	Volume in L
A	1,187.0
B	2,891.0
C	3,068.0
D	1,548.0

Table 4: Optimal distribution of fuel to minimise cost after increasing warehouse C's capacity.

Communication 3

Your request in your third communication was to account for fuel evaporation losses during transport in our optimised solution. The optimised costs of fuel delivery from the four warehouse locations to the burn sites are as follows:

- Purchasing costs: \$76,230.41
- Transport costs: \$903,330.30
- Total costs: \$979,560.70
- Evaporation losses: \$75,849.45

Warehouse	Volume in L
A	1,835.77
B	3,100.00
C	2,100.00
D	2,202.41

Table 5: Optimal distribution of fuel to minimise cost. Warehouses operating at full capacity are shown in red.

Warehouses at maximum capacity

The biggest expense continues to be transportation, representing 92% of the costs. One way to optimise it further is to expand warehouse C's capacity. In this setting (considering road limits and fuel evaporation), increasing it by 1,070 L to a total of 3,170 L, would give the optimal cost:

- Purchasing costs: \$79,153.37
- Transport costs: \$790,881.27
- Total costs: \$916,426.29

This would mean a reduction of \$63,134.41 (6.45%) in total expenditure, particularly impacted by savings in transportation, from \$832,397.04 to \$790,881.27. Distribution of fuel purchases is detailed in Table 6.

Fuel lost to evaporation

The amount of fuel lost due to evaporation was 544.17 L in communication 3 setting and 504.38 L if warehouse C's capacity was expanded. Brolga could save up to \$75,849.45 (7.7%) if evaporation during transport was reduced to 0%.

Warehouse	Volume in L
A	1,248.20
B	3,097.77
C	3,163.92
D	1,688.49

Table 6: Optimal distribution of fuel to minimise cost after increasing warehouse C's capacity.

Roads at maximum capacity

The following roads connecting the sites (45, 41), (4, 51), (5, 37), (5, 23), (37, 35) are operating at maximum capacity. Another way to further optimise costs would be to increase these roads' capacities.

Communication 4

Communication Four introduced the requirement to transport fire suppressant to the burn sites, in addition to burn fuel. The optimised costs of the delivery of fuel and suppressant from the four warehouse locations to the burn sites is shown below:

- Fuel purchasing costs: \$75,809.17
- Suppressant purchasing costs: \$10,889.21
- Transport costs: \$1,221,097.92
- Total costs: \$1,307,796.30

Distribution of fuel and suppressant purchases is detailed in Table 7

Warehouse	Fuel volume (L)	Suppressant volume (L)
A	1,745.98	463.82
B	3,100.00	1,100.00
C	2,100.00	900.00
D	2,300.00	576.18

Table 7: Optimal distribution of fuel and suppressant to minimise cost. Warehouses operating at full capacity are shown in red.

Warehouses at maximum capacity

The biggest expense continues to be transportation, now representing 93% of the costs. One way to optimise it further is to increase the capacity of warehouses that are closer to burn sites. The best result is obtained by the following increases

Warehouse	Fuel capacity (L)	Suppressant capacity (L)
A	2,500	800
B	3,100 → 3,410	1,100 → 1,120
C	2,100 → 2,680	900
D	2,300	1,000

Table 8: Recommended warehouse capacity expansion to obtain optimal cost.

Combining both expansions would result in the costs:

- Fuel purchasing costs: \$77,841.74
- Suppressant purchasing costs: \$10,817.34
- Transport costs: \$1,156,133.32
- Total costs: \$1,244,792.41

This would mean a reduction of \$63,003.89 (4.82%) in total expenditure, particularly impacted by lowering transportation costs from \$832,397.04 to \$790,881.27. Distribution of fuel purchases is detailed in Table 9.

Warehouse	Fuel volume (L)	Suppressant volume (L)
A	1,248.20	640.0
B	3,400.48	1,117.0
C	2,673.00	827.0
D	1,890.17	456.0

Table 9: Optimal distribution of fuel to minimise cost after increasing warehouses' capacities.

Fuel lost to evaporation

The amount of burn fuel lost due to evaporation was 551.98 L in communication 4 setting and 517.85 L if warehouses B and C' capacities were expanded. Brolga could save up to \$94,290.07 (7.2%) if evaporation during transport was reduced to 0%.

Roads at maximum capacity

The roads connecting the sites (45, 41), (4, 51), (4, 46), (29, 43), (43, 25), (45, 53), (53, 42), (5, 37), (5, 23) are operating at maximum capacity. Another way to further optimise costs would be to increase these roads' capacities.

Communication 5

The final communication outlined a five-year forward planning requirement to account for rising transport costs and ensure sufficient storage of fuel and suppressant for future use.

The yearly expenditure in fuel and suppressant purchases, transport costs and their totals are detailed in Table 10, which also shows the amounts of fuel and suppressant that is stored at the end of the season's operations.

Year	Fuel(\$)	Supp.(\$)	Transport(\$)	Fuel(L)	Supp.(L)	Yr Total(\$)
2025	82,826.24	11,365.74	1,435,234.21	646.55	223.00	1,529,426.19
2026	81,187.20	10,856.16	1,413,351.04	953.51	274.00	1,505,394.40
2027	79,168.19	10,850.20	1,422,532.63	1,236.32	300.00	1,512,551.01
2028	75,247.07	11,368.38	1,437,734.11	998.86	565.88	1,524,349.56
2029	69,176.36	9,311.02	1,308,014.28	0.00	0.00	1,386,501.66
Total	387,605.06	53,751.50	7,016,866.27	-	-	7,458,222.82

Table 10: Overview of yearly costs, and yearly totals of stored and transported Fuel and Suppressant (Supp).

The distribution of fuel and suppressant purchases per warehouse per year is detailed in Table 11

Warehouse	Year	Fuel volume (L)	Suppressant volume (L)
A	2025	2,490.90	463.00
	2026	2,316.90	292.00
	2027	2,102.57	290.00
	2028	1,686.31	463.88
	2029	1,041.86	210.12
B	2025	3,100.00	1,100.00
	2026	3,100.00	1,100.00
	2027	3,100.00	1,100.00
	2028	3,100.00	1,100.00
	2029	3,100.00	1,100.00
C	2025	2,100.00	900.00
	2026	2,100.00	900.00
	2027	2,100.00	900.00
	2028	2,100.00	900.00
	2029	2,100.00	900.00
D	2025	2,300.00	800.00
	2026	2,300.00	800.00
	2027	2,300.00	800.00
	2028	2,300.00	800.00
	2029	2,300.00	192.00

Table 11: Optimal distribution of fuel and suppressant to minimise cost for the whole period 2025-2029. Warehouses operating at full capacity are shown in **red**.

Warehouses at maximum capacity

Not surprisingly, the biggest expense is still transportation, now representing 94% of the costs. Therefore, increasing the capacity of warehouses that are closer to burn sites would further optimize the total expenditure. The best result is obtained by the following increases

Warehouse	Fuel capacity (L)	Suppressant capacity (L)
A	2,500	800 → 910
B	3,100 → 3,930	1,100 → 1,610
C	2,100 → 2,740	900 → 940
D	2,300 → 2,510	1,000

Table 12: Recommended warehouse capacity expansion to obtain optimal cost.

Combining these expansions would result in the following costs:

- Fuel purchasing costs: \$396,280.35
- Suppressant purchasing costs: \$54,080.234
- Transport costs: \$6,501,113.47
- Total costs: \$6,951,474.05

This would mean a reduction of \$506,748.77 (6.79%) in total expenditure, particularly driven by lowering transportation costs from \$7,016,866.28 to \$6,501,113.47.

Distribution of fuel purchases with expanded warehouses is detailed in Table 13.

Fuel lost to evaporation

The amount of burn fuel lost due to evaporation was 2,824.54 L in communication 5 setting and 2,604.26 L if warehouses capacities were expanded. Brolga could save up to \$602,687.03 (8.1%) if evaporation during transport was reduced to 0%.

Roads at maximum capacity

Table 14 shows the roads operating at maximum capacity in each year. Another way to further optimise costs would be to increase these roads' capacities.

Warehouse	Year	Fuel volume (L)	Suppressant volume (L)
A	2025	1,995.29	909.83
	2026	1,756.41	640.52
	2027	1,331.70	798.15
	2028	1,092.02	669.88
	2029	344.86	222.12
B	2025	3,465.67	1,607.81
	2026	3,655.94	1,287.50
	2027	3,929.60	981.87
	2028	3,895.65	835.50
	2029	2,592.95	659.00
C	2025	2,567.16	932.84
	2026	2,591.03	908.97
	2027	2,631.00	869.00
	2028	2,727.00	773.00
	2029	2,738.00	762.00
D	2025	2,504.94	456.00
	2026	1,890.79	489.23
	2027	1,924.60	435.77
	2028	1,834.29	557.00
	2029	1,449.36	315.00

Table 13: Optimal distribution of fuel and suppressant to minimise cost for the whole period 2025-2029 after proposed warehouses expansions.

Year	Roads(from site,to site)
2025	(0,7),(45,40),(45,41),(41,30),(48,43),(4,51),(4,46), (29,43),(0,29),(43,25),(45,53),(53,42),(5,37),(5,23)
2026	(0,7),(45,40),(45,41),(41,30),(4,51),(4,46),(29,43), (0,29),(43,25),(45,53),(53,42),(5,37),(5,23)
2027	(45,40),(45,41),(41,30),(4,51),(4,46),(29,43) ,(43,25),(45,53),(53,42),(5,37),(5,23)
2028	(45,40),(45,41),(41,30),(4,51),(4,46),(29,43), (43,25),(45,53),(53,42),(5,37),(5,23)
2029	(45,41),(41,30),(4,51),(4,46),(29,43),(43,25), (45,53),(53,42),(5,37),(5,23)

Table 14: Roads at maximum capacity in communication 5 setting.